

**Project Title**  
**RouteOpt: A Data-Cleaning and Approximation Engine for  
Capacitated Multi-Vehicle Route Optimization with Time  
Windows**

*Project Report submitted for the partial fulfilment of grade for the course*

**Data Structures and Algorithms -3 (25CS2103E)**

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## **ABSTRACT**

Logistics operators must plan daily routes visiting many stops at minimum total distance and time while respecting vehicle capacity and delivery time windows. This problem, a variant of the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), is NP-hard and exactly solvable only for small instances, so real-world fleets require fast approximation methods. This project presents RouteOpt, an engine that ingests raw stop and address data, cleans and geocodes it, and produces near-optimal per-vehicle routes. The pipeline standardizes messy addresses, deduplicates stops, clusters and assigns them to vehicles under capacity constraints, and applies construction heuristics (nearest-neighbor, savings algorithm) followed by local-search improvement (2-opt, Or-opt) to minimize travel cost while honoring time windows. The system reports total distance, estimated travel time, vehicle utilization, and time-window violations for each route, giving operators a measurable trade-off between solution quality and computation time compared to exact solvers.

Signature of the Guide